Digital Mohalla — Component diagram, iteration 1 (design step A8, v1.1): the overview's ten modules; 1, 3, 5, 9 and 10 in detail

Ball = provided interface; dashed arrow = required interface (dependency). Colour = owner (S1 / S2 / S3, Iteration 1 plan work division). Operations of every interface: Architecture_Design.md section 3.

«component»
Mobile app (Expo / React Native)
one app for residents, lenders, moderators and admins (screens W1–W7)

REST API (JSON over HTTPS)

«subsystem» **Digital Mohalla backend** (Python / FastAPI service, PostgreSQL; proposal §8.1, confirmed at the D-15 design gate)

«component»
REST API /api/v1
authentication · authorisation · validation · privacy filter (NFR-PR-01) · errors 403 / 409 / 422 · /health

IAuth

IBorrowingCommands

«component»
Module 1 — Accounts & communities [S1]

AuthService
password login, sessions, roles
(FR-A-04 part; seeded users)

MembershipDirectory
memberships, moderator of community
(seeded in it.1; A3/A4 in it.2)

LocalityVerifier
coarse outcome only: verified | inconclusive | conflicting
A11: geofence + invitation; «stub» until the D-11 gate decides the build

IMembership

ILocality

«component»
Module 3 — Borrowing lifecycle [S1]

BorrowingService
command handlers B5–B8, B12

TransitionValidator
A3 transition table + authorisation

EventStore
append-only, stream_id + stream_seq

Projector & Replay
current state = fold(events)

BookingGuard
EXCLUDE constraint (ADR-0002)

OutcomePublisher
closing events → IOutcomeSink

IOutcomeSink

«component»
Module 5 — Trust, reputation & safety (assurance engine) [S2 + S1]

AssuranceFacade
assess(request) pipeline, timeout

IIndirect-Evidence

IndirectTrust [S1]
TrustEdgeProjector, KHopPathSearch (k from config), noisy-OR, ρ ≤ 0.6

BetaPosteriorEstimator
recency-weighted, 90% interval

OutcomeExtractor
confirmed outcomes only

ManipulationDetector
reciprocal clustering, retaliation

data: assessment, outcome, trust_edge, review tables (schema.sql)

ConfigLoader
versioned parameters, config_version
(NFR-D-01)

DecisionPolicy & ExplanationBuilder
assurance state, safeguards

ILifecycleCommands

IEventQuery

IAssess: assess(request) → AssessmentResult

ICaseIntake

INotify

«component»
Module 9 — Administration & moderation [S3]

CaseService
EscalationCase lifecycle (A3), queue, routing to Admin

EvidenceTrailBuilder
trail from the event log (D6)

DecisionService
I4: decision + reason → lifecycle command

AuditLogService
hash-chained log, filters, masking (I6)

Appeal and reversal: iteration 2.

IModeration

«component»
Module 7 — Communication & notifications

notify(user, type, payload)
«stub» in it.1 (log only);
in-app notifications it.2;
chat thread (G1) it.3

«component»
Module 10 — Analytics & evaluation [S3]

ScenarioSuite
ES-01–ES-06, ES-13 fixtures; runs in CI

Frozen event logging (FR-N-03) is the Module 3 event store.

Module 2 — Community events
stretch, it.3 at the earliest

Module 4 — Search, discovery & matching
supporting, it.3

Module 6 — Forum & chatbot
stretch, it.3 at the earliest

Module 8 — Vendors
stretch, it.3 at the earliest

«use» SQL (psycopg)

«component»
PostgreSQL 16 (schema dm, external)
event store, projections, bookings (EXCLUDE), assessments, outcomes, trust edges, cases, audit log — 06_data_model/schema.sql v2.3

Legend
S1 lifecycle, indirect trust, locality S2 estimation, policy, detection S3 moderation, evaluation, apps
shared API and notification stub deferred not in iteration 1
○— provided interface (ball) - - -> required interface (dependency)
The S1/S2 boundary is **IAssess** (Module 3 → Module 5) and, inside Module 5, IIndirectEvidence.